

*Unconformity*

# Unconformity

- It is one of the most common geological feature found in rocks or in succession.
- It is different then all other geological structures viz. the fold, joints and faults
- Unconformities are resulted due to tectonic activity in form of uplift or subsidence of land
- It is referred to a period of non-deposition

→ How unconformities occur?

- The fundamental "laws" of stratigraphy, formulated in the 17th Century by Nicolas Steno, is the law of Original Horizontality, which is known as *Conformity*

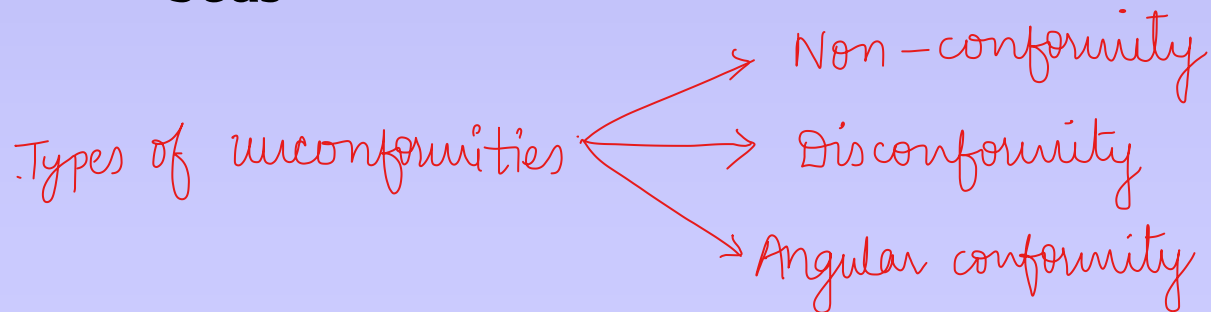
- That is, any deposition when takes place is totally in horizontal fashion

↳ Law of Original Horizontality

- Later due to tectonic movement the layers or beds are **tilted** (*except in case of cross-bedding- which are formed under fluvial (riverine) or aeolian (wind) environment.*)

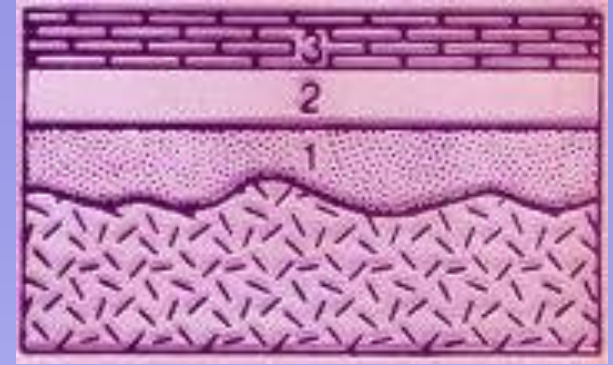
# Reasons for Unconformities

- Formation of unconformity involves:
- ✓ • Horizontal or conformable strata or beds are formed
- ✓ • Break in sedimentation or deposition
- ✓ • Happens due to tectonic movements, that causes uplift or subsidence of land surface
- ✓ • Next phase of Deposition or sedimentation cycle, where new sedimentation produce another set of conformable beds



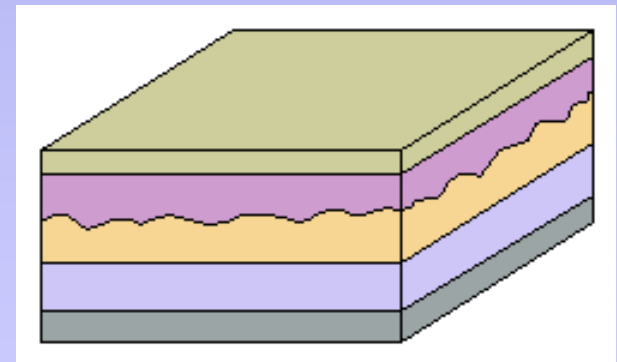
# Types of Unconformities

- **Non-conformity**
- When the underlying rocks are Igneous or Metamorphic (i.e. unstratified) and the overlying younger rocks are sedimentary (stratified) = Non-conformity
- **Disconformity**
- When the underlying (older) and overlying (younger) sedimentary rock strata are parallel and the contact plane is an erosional surface = Disconformity



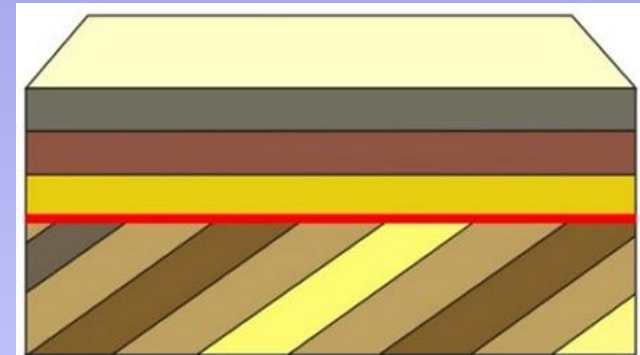
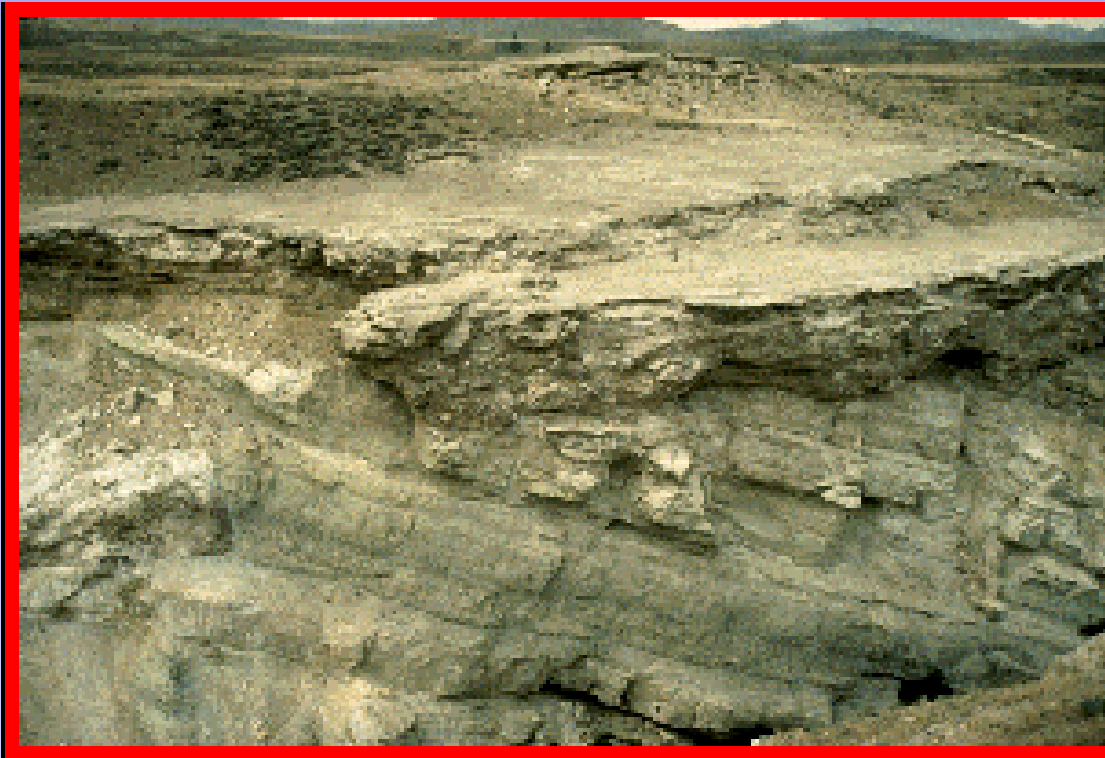
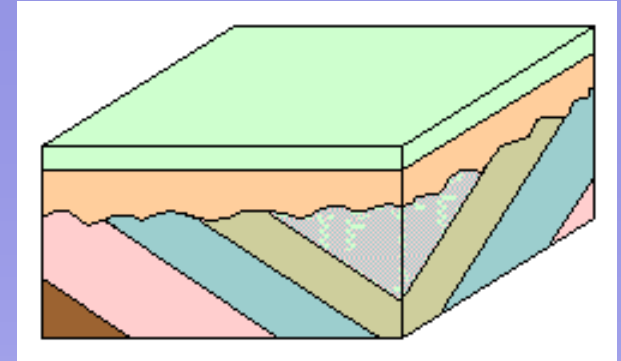
→ younger rock  
sedimentary (stratified)

Igneous/metamorphic  
(unstratified)

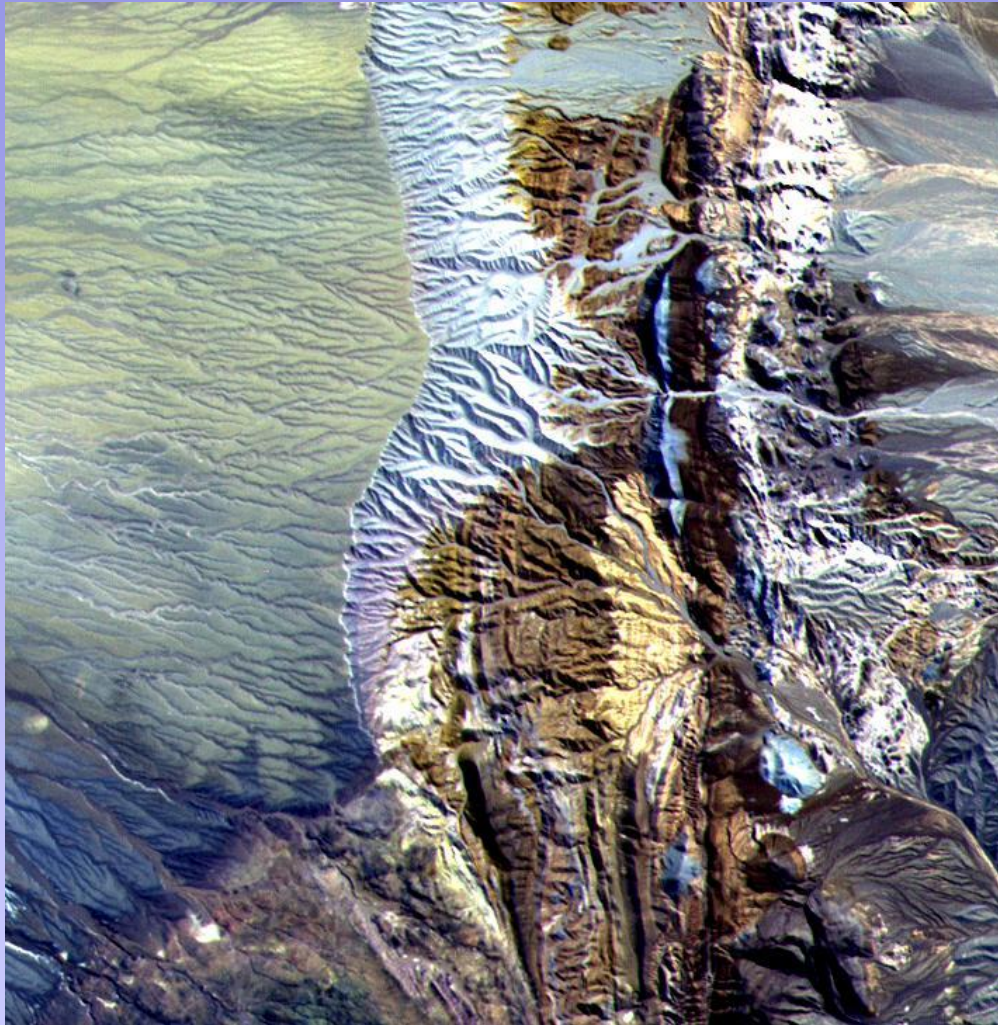


sedimentary (younger)  
~~sedimentary~~  
(older) → sedimentary

- **Angular unconformity**
- When the underlying (older) rocks and overlying (younger) rock strata show some angle w.r.t one another=Angular unconformity



# Angular unconformity



- This sub-area in northern Chile Showing a geological angular unconformity: a contact between layers of rock at different angles.
- On the right side of the image, Cretaceous sediments were tilted upward to an angle of about 50 degrees, then eroded. On this surface volcanic pyroclastic deposits were deposited as a flat sheet.
- The section of rocks has been eroding from the east, exposing the tilted and flat rock layers.



